Qinhe Peng

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EDUCATION

Tsinghua University — Sep. 2020 – Jun. 2025
Bachelor of Engineering — *GPA: 3.7/4.0*
Electronic Information Science and Technology
Thesis: Robot Planning and Control Based on Deep Learning
Advised by Yi Wu, Thesis Defense Grade: 4.0/4.0

University of Pennsylvania — Sep. 2025 – Jun. 2027
Master of Engineering — *GPA: 3.6/4.0*
Robotics

RESEARCH INTERESTS

Mobile and Dexterous Manipulation;
Learning from Videos and Cross-Embodiment Transfer;
Agentic and Efficient Robotics in Real World Deployment;
Physical AI Safety through Control and Interpretability.

PUBLICATIONS

Words Are Not Motions: Steering Vision-Language-Action Models with Jacobian Pullbacks
CORL2026 workshop, RSS 2026 submitting
Qinhe Peng, Dinesh Jayaraman, Subin Kim.
[paper] [website]

π-IMLE: Fast Single-Step Action Generation for Vision-Language-Action Policies
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
Kian Hosseinkhani, Qinhe Peng, George Shramko, Mehran Aghabozorgi, Jianing Qian, Tristan Engst, Alireza Moazeni, Dinesh Jayaraman, Ke Li.
[paper] [website] [code]

Scaling Spatial and Temporal Context for Robotic Imitation Learning Policies With Scene Graphs
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
Jianing Qian, Qinhe Peng, Emmanuel Panov, Leonor Fermoselle, Dinesh Jayaraman, Bernadette Bucher, Tarik Kelestemur.
[paper] [website] [code]

JetViT: Efficient High-Resolution Vision Transformer with Post-Training Attention Search
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
Dongyun Zou, Zhuoyang Zhang, Junyu Chen, Wenkun He, Qinhe Peng, Hanrong Ye, Yao Lu, Hongxu Yin, Yu Wang, Song Han, Han Cai.
[paper]

A Closed-Loop Control System with Ultrafast Monitoring Based on a Photoelectric Sensor
IEEE SENSORS, 2024, pp. 1–4. ***Oral presentation***
Yang Yuanming, Qinhe Peng, Xinxing Yuan, Panyu Hou, Binxiang Qi, Luming Duan.
[paper]

RESEARCH EXPERIENCE

Jacobian Interpreting and Steering VLA Jul. – Oct. 2026
GRASP Lab, University of Pennsylvania
Advisor: Prof. Dinesh Jayaraman

- Developed a generalizable framework using Jacobian pullbacks to map semantic and kinematic objectives into activation steering directions without modifying policy weights or training auxiliary models.
- Designed contrastive residual interventions to steer motion behaviour and object selection across OpenVLA and $\pi_{0.5}$ in simulation and real-world manipulation, with a 38% increase in speed and up to 27% lower motion roughness.
- Characterized the distinction between semantic and motor representations, showing that combining semantic and kinematic directions improves behavioral control.

Fast Single-Step Action Generation for VLA Policies Feb. – Aug. 2026
GRASP Lab, University of Pennsylvania
Advisor: Prof. Dinesh Jayaraman, Prof. Ke Li

- Replaced iterative flow-matching action generation with a single-step conditional generator trained through conditional Implicit Maximum Likelihood Estimation.
- Preserved multimodal action coverage while increasing $\pi_{0.5}$ inference frequency from 15 Hz to 55 Hz (3.67×), halving jerk and cumulative VLA inference time.
- Achieved higher (98.0%) average success while retaining robustness under test-time perturbations.

Scene Graph Enabled Robot Imitation Learning Sep. – Nov. 2025
GRASP Lab, University of Pennsylvania
Advisor: Prof. Dinesh Jayaraman

- Developed dynamic, object-centric scene graph memory for imitation learning in large-scale environments with partial and changing observability.
- Used scene information to support long-horizon, multi-step robotic tasks.
- Improved Diffusion Policy performance in simulated mobile manipulation and real-world tabletop manipulation, with stronger generalization and robustness under severe partial observability.

Efficient Medical Image and Video Segmentation Dec. 2023 – Aug. 2026
MIT Han Lab
Advisor: Prof. Song Han

- Developed an EfficientViT-based medical image segmentation model for real-time analysis in resource-constrained environments.
- Halved multiply–accumulate operations (MACs), reduced parameters to less than one-eighth of the original model, and more than doubled throughput.
- Outperformed the previous leading model on validation and zero-shot datasets using the same fine-tuning data, with reported Dice score improvements of 5–29% across five zero-shot tests.
- Extended efficient segmentation to video and concept segmentation using SAM2 and SAM3.

High-Precision PID Laser Signal Control Sep. 2023 – Dec. 2024
Center for Quantum Information, Tsinghua University
Advisor: Prof. Luming Duan

- Applied neural network algorithms to stabilize laser frequency and amplitude in high-precision control systems.
- Simulated PID algorithms in MATLAB and performed experiments integrating optical systems, FPGA hardware, and PID control.
- Implemented PC-side UDP communication, waveform visualization, and frequency analysis, together with an FPGA-based arbitrary waveform generator.
- Led a project funded by the Beijing Natural Science Foundation; related work was accepted for an oral presentation at IEEE SENSORS 2024.

Robust Design of On-Chip Optical Systems Nov. 2022 – Mar. 2023
Tsinghua University
Advisor: Research Scientist Jiyuan Zheng

- Applied neural network generalization principles to optical chip design to improve robustness to fabrication errors, including in diffractive neural network adder chips.
- Developed a robust design method and reusable implementation with interfaces for different application scenarios.

HONORS & AWARDS

Beijing Natural Science Foundation Funding 2025

BiRen Technology GPU Programming Competition Winner 2024

Science and Technology Innovation Excellent Scholarship 2023, 2024
Tsinghua University

Zheng Gang Scholarship 2023
Tsinghua University

Student Research Training Outstanding Project Award 2022
Tsinghua University; Top 2% of whole university

Innovation Talent Training Program 2022
Tsinghua University, Electrical Engineering Department

Tsinghua University Scholarship 2022
Tsinghua University

TEACHING & SERVING

Applied Machine Learning, Teaching Assistant Spring 2026
University of Pennsylvania

Data and Algorithm, Teaching Assistant Fall 2023, Spring 2024
Tsinghua University

CORL2026, Reviewer

Student Association of Science and Technology, Organizer 2023, 2024
Tsinghua University, Electrical Engineering Department

Artificial Intelligence Challenge, Organizer 2023, 2024
Tsinghua University

SKILLS

Languages: Chinese (Native), English (Fluent). TOEFL: 111; GRE: 331(3.5)

Software and Development: Python, C/C++, C#, Bash, MATLAB/Simulink, PyTorch, JAX, OpenCV, Linux, Git, Docker.

Robotics and Simulation: ROS/ROS 2, MuJoCo, Isaac Sim, ManiSkill, DROID.

Hardware and Engineering: Verilog, VHDL, FPGA, Arduino, Vivado, Quartus, SolidWorks, UART, UDP.

RELATED COURSEWORK

Tsinghua University

Intelligent Systems and Robotics
Data and Algorithm
Embodied Artificial Intelligence
Probability and Stochastic Processes
Intelligent UAV Design and Implementation
Electronic Circuits and System
Digital Logic and processor
Signals and Systems
Operating System
Statistical Signal Processing

University of Pennsylvania

Engineering Mathematics
Artificial Intelligence
Feedback Control
Machine Perception
Deep Generative Models